

MSS_Assignment I – FIO

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【實驗設備說明】

- 測試平台：MSI GF65 Thin 筆電
- 儲存裝置：Samsung PM981a NVMe SSD
- 作業系統：Ubuntu 22.04
- fio 版本：3.28

【Q0】

Please run q0.fio on your own device and write down your output q0 in report.

■ 測試方式：

使用 fio 對一個 64MB 檔案 `testfile` 做順序讀取，區塊大小為 4KB。
檔案儲存在系統的 NVMe SSD (/dev/nvme0n1) 上。

■ 測試結果摘要：

IOPS：55.9k

Bandwidth：218MiB/s (229MB/s)

Lat(avg)：17.3μs

```
shihcw@shihcw-GF65-Thin-9SD:~/Work_space/MSS$ cat output.txt
job1: (g=0): rw=read, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
fio-3.28
Starting 1 process
job1: Laying out IO file (1 file / 64MiB)

job1: (groupid=0, jobs=1): err= 0: pid=302080: Fri Apr 11 15:51:11 2025
read: IOPS=55.9k, BW=218MiB/s (229MB/s)(64.0MiB/293msec)
   clat (nsec): min=13517, max=98622, avg=17271.33, stdev=4657.13
   lat (nsec): min=13552, max=98912, avg=17312.38, stdev=4667.70
   clat percentiles (nsec):
    |  1.00th=[14272],  5.00th=[14528], 10.00th=[14528], 20.00th=[14656],
    | 30.00th=[14784], 40.00th=[15168], 50.00th=[16512], 60.00th=[16768],
    | 70.00th=[17280], 80.00th=[17792], 90.00th=[22144], 95.00th=[23936],
    | 99.00th=[37632], 99.50th=[46848], 99.90th=[66048], 99.95th=[74240],
    | 99.99th=[83456]
   lat (usec)   : 20=86.02%, 50=13.57%, 100=0.42%
   cpu          : usr=5.14%, sys=29.45%, ctx=16385, majf=0, minf=14
IO depths      : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
   submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
   complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
   issued rwts: total=16384,0,0,0 short=0,0,0,0 dropped=0,0,0,0
   latency     : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
   READ: bw=218MiB/s (229MB/s), 218MiB/s-218MiB/s (229MB/s-229MB/s), io=64.0MiB (67.1MB), run=293-293ms

Disk stats (read/write):
nvme0n1: ios=14698/0, merge=1/0, ticks=189/0, in_queue=189, util=53.15%
```

【Q1】

Is there any significant difference between read and randread? Why or why not? Please justify your answer in brief.

■ 測試結果摘要：

[Read]

- IOPS：57.9k
- Bandwidth：226MiB/s
- Lat(avg)：16.96 μ s

[Randread]

- IOPS：18.2k
- Bandwidth：71.0MiB/s
- Lat(avg)：54.54 μ s

■ 分析：

順序讀取的效能明顯優於隨機讀取，主要原因如下：

1. 順序讀取資料是連續的，作業系統與儲存裝置能有效進行預讀，降低磁碟存取延遲。
2. 隨機讀取會導致大量尋址與跳躍式存取，使得快取命中率下降、延遲提高，特別是在讀取大量小區塊（4KB）時更明顯。
3. 實測中，順序讀取的 IOPS 與 Bandwidth 明顯高於隨機讀，平均延遲也顯著較低，符合預期的效能特性。

```

shihcw@shihcw-GF65-Thin-9SD:~/Work_space/HSS/FIO/Q_1$ cat output_q1.txt
read: (g=0): rw=read, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
randread: (g=1): rw=randread, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
fio-3.28
Starting 2 processes
read: Laying out IO file (1 file / 64MiB)

read: (groupid=0, jobs=1): err= 0: pid=302876: Fri Apr 11 16:09:52 2025
  read: IOPS=57.9k, BW=226MiB/s (237MB/s)(64.0MiB/283msec)
    clat (usec): min=12, max=1102, avg=16.94, stdev=17.05
    lat (usec): min=12, max=1102, avg=16.96, stdev=17.06
    clat percentiles (usec):
      | 1.00th=[ 14], 5.00th=[ 15], 10.00th=[ 15], 20.00th=[ 15],
      | 30.00th=[ 15], 40.00th=[ 15], 50.00th=[ 16], 60.00th=[ 17],
      | 70.00th=[ 17], 80.00th=[ 18], 90.00th=[ 19], 95.00th=[ 23],
      | 99.00th=[ 37], 99.50th=[ 46], 99.90th=[ 102], 99.95th=[ 115],
      | 99.99th=[ 1074]
    lat (usec)   : 20=92.29%, 50=7.29%, 100=0.30%, 250=0.09%, 500=0.01%
    lat (usec)   : 1000=0.01%
    lat (msec)   : 2=0.02%
    cpu          : usr=5.67%, sys=27.30%, ctx=16382, majf=0, minf=14
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
      submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      issued rwts: total=16384,0,0,0 short=0,0,0,0 dropped=0,0,0,0
      latency    : target=0, window=0, percentile=100.00%, depth=1
randread: (groupid=1, jobs=1): err= 0: pid=302877: Fri Apr 11 16:09:52 2025
  read: IOPS=18.2k, BW=71.0MiB/s (74.4MB/s)(64.0MiB/902msec)
    clat (usec): min=49, max=164, avg=54.51, stdev= 6.01
    lat (usec): min=49, max=164, avg=54.54, stdev= 6.02
    clat percentiles (usec):
      | 1.00th=[ 51], 5.00th=[ 51], 10.00th=[ 51], 20.00th=[ 52],
      | 30.00th=[ 52], 40.00th=[ 53], 50.00th=[ 53], 60.00th=[ 54],
      | 70.00th=[ 55], 80.00th=[ 56], 90.00th=[ 60], 95.00th=[ 63],
      | 99.00th=[ 83], 99.50th=[ 96], 99.90th=[ 112], 99.95th=[ 130],
      | 99.99th=[ 153]
    bw ( KiB/s): min=73000, max=73000, per=100.00%, avg=73000.00, stdev= 0.00, samples=1
    iops        : min=18250, max=18250, avg=18250.00, stdev= 0.00, samples=1
    lat (usec)   : 50=0.01%, 100=99.76%, 250=0.23%
    cpu          : usr=2.66%, sys=9.10%, ctx=16384, majf=0, minf=12
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
      submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      issued rwts: total=16384,0,0,0 short=0,0,0,0 dropped=0,0,0,0
      latency    : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=226MiB/s (237MB/s), 226MiB/s-226MiB/s (237MB/s-237MB/s), io=64.0MiB (67.1MB), run=283-283msec

Run status group 1 (all jobs):
  READ: bw=71.0MiB/s (74.4MB/s), 71.0MiB/s-71.0MiB/s (74.4MB/s-74.4MB/s), io=64.0MiB (67.1MB), run=902-902msec

Disk stats (read/write):
nvme0n1: ios=30783/60, merge=2/18, ticks=920/22, in queue=946, util=66.59%

```

【Q2】

Is there any significant difference between write and randwrite? Why or why not? Please justify your answer in brief.

■ 測試結果摘要：

[write]

- IOPS：33.9k
- Bandwidth：132MiB/s
- Lat(avg)：29.03 μ s

[randwrite]

- IOPS：45.5k
- Bandwidth：178MiB/s
- Lat(avg)：21.42 μ s

■ 分析：

在這次實驗中，隨機寫入效能反而高於順序寫入，這個結果與傳統磁碟（HDD）常見的特性相反。

原因可能包括：

1. 這顆 NVMe SSD 有非常好的寫入控制器與快取策略，能快速分配和處理隨機寫入請求。
2. 寫入量不大（64MB），尚未產生寫入放大效應（Write Amplification）或垃圾回收（GC）壓力。
3. 實驗為模擬檔案寫入，受限於檔案系統行為，不代表實際對裸裝置的效能。

儘管結果中隨機寫入表現較好，但在真實長時間高寫入壓力下，順序寫入仍通常更穩定、延遲更低。

```

shihcw@shihcw-GF65-Thin-9SD:~/Work_space/HSS/FIO/Q_2$ cat output_q2.txt
write: (g=0): rw=write, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
randwrite: (g=1): rw=randwrite, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
fio-3.28
Starting 2 processes
write: Laying out IO file (1 file / 64MiB)

write: (groupid=0, jobs=1): err= 0: pid=303094: Fri Apr 11 16:14:44 2025
  write: IOPS=33.9k, BW=132MiB/s (139MB/s)(64.0MiB/484msec); 0 zone resets
    clat (usec): min=22, max=8264, avg=28.96, stdev=66.47
    lat (usec): min=22, max=8264, avg=29.03, stdev=66.48
    clat percentiles (usec):
      | 1.00th=[ 26], 5.00th=[ 26], 10.00th=[ 27], 20.00th=[ 27],
      | 30.00th=[ 28], 40.00th=[ 28], 50.00th=[ 28], 60.00th=[ 28],
      | 70.00th=[ 28], 80.00th=[ 29], 90.00th=[ 32], 95.00th=[ 35],
      | 99.00th=[ 49], 99.50th=[ 59], 99.90th=[ 88], 99.95th=[ 96],
      | 99.99th=[ 2073]
    lat (usec)   : 50=99.10%, 100=0.85%, 250=0.04%
    lat (msec)   : 4=0.01%, 10=0.01%
    cpu          : usr=4.14%, sys=34.16%, ctx=16385, majf=0, minf=13
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
      submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      issued rwts: total=0,16384,0,0 short=0,0,0,0 dropped=0,0,0,0
      latency    : target=0, window=0, percentile=100.00%, depth=1
randwrite: (groupid=1, jobs=1): err= 0: pid=303095: Fri Apr 11 16:14:44 2025
  write: IOPS=45.5k, BW=178MiB/s (186MB/s)(64.0MiB/360msec); 0 zone resets
    clat (usec): min=16, max=135, avg=21.33, stdev= 6.27
    lat (usec): min=16, max=163, avg=21.42, stdev= 6.36
    clat percentiles (usec):
      | 1.00th=[ 18], 5.00th=[ 19], 10.00th=[ 19], 20.00th=[ 19],
      | 30.00th=[ 20], 40.00th=[ 20], 50.00th=[ 20], 60.00th=[ 20],
      | 70.00th=[ 20], 80.00th=[ 23], 90.00th=[ 27], 95.00th=[ 31],
      | 99.00th=[ 49], 99.50th=[ 59], 99.90th=[ 86], 99.95th=[ 98],
      | 99.99th=[ 110]
    lat (usec)   : 20=73.99%, 50=25.09%, 100=0.89%, 250=0.04%
    cpu          : usr=6.13%, sys=28.97%, ctx=16384, majf=0, minf=11
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
      submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      issued rwts: total=0,16384,0,0 short=0,0,0,0 dropped=0,0,0,0
      latency    : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  WRITE: bw=132MiB/s (139MB/s), 132MiB/s-132MiB/s (139MB/s-139MB/s), io=64.0MiB (67.1MB), run=484-484msec

Run status group 1 (all jobs):
  WRITE: bw=178MiB/s (186MB/s), 178MiB/s-178MiB/s (186MB/s-186MB/s), io=64.0MiB (67.1MB), run=360-360msec

Disk stats (read/write):
  nvme0n1: ios=1/24995, merge=0/15, ticks=0/469, in_queue=471, util=47.64%

```

【Q3】

Is there any significant difference between forward and backward write? Why or why not? Please justify your answer in brief.

■ 實驗方法：

由於無法直接先寫入 0 在寫入 10 到 1，因此透過 `fio` 搭配 `write\_iolog` 模擬兩種區塊寫入順序：

Forward：順序連續寫入

Backward：先寫 block 0，再從檔案尾端向前跳躍式寫入(請參考 backward.log)

■ 測試結果摘要：

[Forward write]

- IOPS：282
- Bandwidth：1128 KiB/s
- Lat(avg)：773.71 μ s

[Backward write]

- IOPS：268
- Bandwidth：1073 KiB/s
- Lat(avg)：764.10 μ s

■ 分析：

從數據來看，forward 與 backward 寫入在本次實驗下效能差異不大，兩者皆為線性單工小量寫入操作。

不過，backward write 的 fsync 延遲略高，可能是因為：

- 寫入區塊不連續，導致較多快取同步等待
- 寫入順序跳躍，讓磁碟控制器的寫入優化策略（如合併、平行）無法發揮
- 雖然是 SSD，但在小區塊跳寫時仍有空間管理與對齊的成本

總體而言，forward write 較穩定、效率稍佳，這也符合作業系統和硬體設計對順序 I/O 的優化特性。

```

shihcw@shihcw-GF65-Thin-950:~/Work_space/HSS/FIO/q3$ cat output_q3.txt
forward: (groupid=0, jobs=1): rw=write, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
backward: (g=1): rw=write, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
fio-3.28
Starting 2 processes
forward: Laying out IO file (1 file / 0MiB)

forward: (groupid=0, jobs=1): err= 0: pid=305124: Fri Apr 11 17:01:29 2025
write: IOPS=282, BW=1128KiB/s (1155kB/s)(44.0KiB/39msec); 0 zone resets
  clat (usec): min=80, max=7645, avg=773.34, stdev=2279.39
  lat (usec): min=80, max=7646, avg=773.71, stdev=2279.38
  clat percentiles (usec):
    | 1.00th=[ 81], 5.00th=[ 81], 10.00th=[ 81], 20.00th=[ 83],
    | 30.00th=[ 83], 40.00th=[ 84], 50.00th=[ 84], 60.00th=[ 88],
    | 70.00th=[ 91], 80.00th=[ 95], 90.00th=[ 96], 95.00th=[ 7635],
    | 99.00th=[ 7635], 99.50th=[ 7635], 99.90th=[ 7635], 99.95th=[ 7635],
    | 99.99th=[ 7635]
  lat (usec)   : 100=90.91%
  lat (msec)   : 10=9.09%
fsync/fdatasync/sync_file_range:
  sync (usec): min=2911, max=3292, avg=3036.19, stdev=110.16
  sync percentiles (usec):
    | 1.00th=[ 2900], 5.00th=[ 2900], 10.00th=[ 2900], 20.00th=[ 2966],
    | 30.00th=[ 2966], 40.00th=[ 2966], 50.00th=[ 2999], 60.00th=[ 2999],
    | 70.00th=[ 3064], 80.00th=[ 3064], 90.00th=[ 3130], 95.00th=[ 3294],
    | 99.00th=[ 3294], 99.50th=[ 3294], 99.90th=[ 3294], 99.95th=[ 3294],
    | 99.99th=[ 3294]
cpu           : usr=0.00%, sys=0.00%, ctx=21, majf=0, minf=14
IO depths     : 1=190.9%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit       : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwts: total=0,11,0,10 short=0,0,0,0 dropped=0,0,0,0
latency      : target=0, window=0, percentile=100.00%, depth=1

backward: (groupid=1, jobs=1): err= 0: pid=305125: Fri Apr 11 17:01:29 2025
write: IOPS=268, BW=1073KiB/s (1099kB/s)(44.0KiB/41msec); 0 zone resets
  clat (usec): min=39, max=7598, avg=762.42, stdev=2267.34
  lat (usec): min=41, max=7603, avg=764.10, stdev=2268.29
  clat percentiles (usec):
    | 1.00th=[ 41], 5.00th=[ 41], 10.00th=[ 80], 20.00th=[ 82],
    | 30.00th=[ 82], 40.00th=[ 82], 50.00th=[ 82], 60.00th=[ 84],
    | 70.00th=[ 87], 80.00th=[ 87], 90.00th=[ 87], 95.00th=[ 7570],
    | 99.00th=[ 7570], 99.50th=[ 7570], 99.90th=[ 7570], 99.95th=[ 7570],
    | 99.99th=[ 7570]
  lat (usec)   : 50=9.09%, 100=81.82%
  lat (msec)   : 10=9.09%
fsync/fdatasync/sync_file_range:
  sync (usec): min=2765, max=5810, avg=3210.79, stdev=917.35
  sync percentiles (usec):
    | 1.00th=[ 2769], 5.00th=[ 2769], 10.00th=[ 2769], 20.00th=[ 2868],
    | 30.00th=[ 2868], 40.00th=[ 2933], 50.00th=[ 2933], 60.00th=[ 2933],
    | 70.00th=[ 2933], 80.00th=[ 2966], 90.00th=[ 3097], 95.00th=[ 5800],
    | 99.00th=[ 5800], 99.50th=[ 5800], 99.90th=[ 5800], 99.95th=[ 5800],
    | 99.99th=[ 5800]
cpu           : usr=0.00%, sys=0.00%, ctx=21, majf=0, minf=13
IO depths     : 1=190.9%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit       : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwts: total=0,11,0,10 short=0,0,0,0 dropped=0,0,0,0
latency      : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  WRITE: bw=1128KiB/s (1155kB/s), 1128KiB/s-1128KiB/s (1155kB/s-1155kB/s), io=44.0KiB (45.1kB), run=39-39msec

Run status group 1 (all jobs):
  WRITE: bw=1073KiB/s (1099kB/s), 1073KiB/s-1073KiB/s (1099kB/s-1099kB/s), io=44.0KiB (45.1kB), run=41-41msec

Disk stats (read/write):
nvme0n1: ios=0/0, merge=0/0, ticks=0/0, in_queue=0, util=0.00%

```

【Q4】

1. (10%) Is there any significant difference between buffered and nonbuffered sequential read? Why or why not? Please justify your answer in brief.
2. (5%) Replace sequential read with random read. Is there any significant difference between buffered and nonbuffered random read? Why or why not? Please justify your answer in brief.

1. Q4-1 : Sequential Read (順序讀取)

■ 測試結果摘要：

[Buffered Sequential Read]

- IOPS : 381k
- Bandwidth : 1488 MiB/s
- Lat(avg) : 2.37 μ s

[Non-buffered Sequential Read]

- IOPS : 56.9k
- Bandwidth : 222 MiB/s
- Lat(avg) : 16.81 μ s

■ 分析：

Buffered 模式透過 OS 的 page cache 預先快取整塊檔案，讓讀取幾乎不經過實體磁碟，因此 IOPS 與 bandwidth 遠高於 non-buffered 模式，延遲也顯著更低。Non-buffered 模式必須直接從磁碟存取資料，雖然硬體為 NVMe SSD，效能仍明顯低於使用 cache 的情況，這證明了快取系統對順序讀效能的巨大貢獻。

2. Q4-2 : Random Read (隨機讀取)

■ 測試結果摘要：

[Buffered Random Read]

- IOPS : 16.0k
- Bandwidth : 62.6 MiB/s
- Lat(avg) : 61.43 μ s

[Non-buffered Random Read]

- IOPS : 14.0k
- Bandwidth : 54.8 MiB/s
- Lat(avg) : 68.61 μ s

■ 分析：

隨機讀取本身具有大量的跳躍與快取 miss 機會，即使啟用快取，也難以像順序讀取那樣預先抓取有用資料。因此 Buffered 與 Non-buffered 模式下的效能差距比順序讀小很多。不過還是可以看到 buffered 模式仍略快，這可能是因為某些 block 被重複讀取、或系統有部分預讀行為。

3. Q4-2 :

Is there any significant difference between buffered and nonbuffered random read? Why or why not? Please justify your answer in brief.

Ans :

buffered 與 nonbuffered 隨機讀取之間存在一定差異，但差異不如順序讀取明顯。因為 buffered 模式會使用作業系統的 page cache，若讀取位置有重複或可被預先快取，效能會略好，延遲也較低。nonbuffered 模式完全繞過快取，每次讀取都要直接從儲存裝置讀取資料，因此效能通常較差，特別是在大量隨機存取時。然而，由於隨機讀的存取位置分散，即使是 buffered 模式，快取命中率也偏低，因此整體效能提升有限，差距不如順序讀明顯。

```
shihcw@shihcw-GF65-Thin-95D:~/Work_space/HSS/FIO/q4$ cat output_q4.txt
buffered_seqread: (g=0): rw=read, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
nonbuffered_seqread: (g=1): rw=read, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
buffered_randread: (g=2): rw=randread, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
nonbuffered_randread: (g=3): rw=randread, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
fio-3.28
Starting 4 processes
buffered_seqread: Laying out IO file (1 file / 64MiB)

buffered_seqread: (groupid=0, jobs=1): err= 0: pid=305853: Fri Apr 11 17:07:11 2025
read: IOPS=381k, BW=1488MiB/s (1561MB/s)(64.0MiB/43msec)
   clat (nsec): min=1148, max=139933, avg=2343.36, stdev=7518.05
   lat (nsec): min=1172, max=140240, avg=2370.14, stdev=7518.41
  clat percentiles (nsec):
    |  1.00th=[ 1160],  5.00th=[ 1160], 10.00th=[ 1176], 20.00th=[ 1176],
    | 30.00th=[ 1176], 40.00th=[ 1192], 50.00th=[ 1208], 60.00th=[ 1224],
    | 70.00th=[ 1320], 80.00th=[ 1464], 90.00th=[ 1800], 95.00th=[ 3312],
    | 99.00th=[ 55552], 99.50th=[ 60672], 99.90th=[ 82432], 99.95th=[ 95744],
    | 99.99th=[128512]
  lat (usec)   : 2=91.24%, 4=6.21%, 10=0.95%, 20=0.03%, 50=0.20%
  lat (usec)   : 100=1.35%, 250=0.02%
  cpu          : usr=23.81%, sys=71.43%, ctx=3, majf=0, minf=12
 IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
 submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
 complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
 issued rwts: total=16384,0,0,0 short=0,0,0,0 dropped=0,0,0,0
 latency     : target=0, window=0, percentile=100.00%, depth=1

nonbuffered_seqread: (groupid=1, jobs=1): err= 0: pid=305854: Fri Apr 11 17:07:11 2025
read: IOPS=56.9k, BW=222MiB/s (233MB/s)(64.0MiB/288msec)
   clat (usec): min=13, max=444, avg=16.78, stdev= 5.72
   lat (usec): min=13, max=444, avg=16.81, stdev= 5.73
  clat percentiles (usec):
    |  1.00th=[ 15],  5.00th=[ 15], 10.00th=[ 15], 20.00th=[ 15],
    | 30.00th=[ 15], 40.00th=[ 16], 50.00th=[ 16], 60.00th=[ 17],
    | 70.00th=[ 18], 80.00th=[ 18], 90.00th=[ 20], 95.00th=[ 23],
    | 99.00th=[ 35], 99.50th=[ 45], 99.90th=[ 78], 99.95th=[ 88],
    | 99.99th=[ 143]
  lat (usec)   : 20=91.74%, 50=7.87%, 100=0.37%, 250=0.02%, 500=0.01%
  cpu          : usr=6.97%, sys=28.92%, ctx=16383, majf=0, minf=13
 IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
 submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
 complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
 issued rwts: total=16384,0,0,0 short=0,0,0,0 dropped=0,0,0,0
 latency     : target=0, window=0, percentile=100.00%, depth=1
```

```

buffered_randread: (groupid=2, jobs=1): err= 0: pid=305855: Fri Apr 11 17:07:11 2025
read: IOPS=16.0k, BW=62.6MiB/s (65.7MB/s)(64.0MiB/1022msec)
  clat (usec): min=50, max=702, avg=61.34, stdev=13.73
    lat (usec): min=51, max=702, avg=61.43, stdev=13.77
  clat percentiles (usec):
    | 1.00th=[ 52], 5.00th=[ 52], 10.00th=[ 53], 20.00th=[ 53],
    | 30.00th=[ 55], 40.00th=[ 56], 50.00th=[ 58], 60.00th=[ 60],
    | 70.00th=[ 63], 80.00th=[ 68], 90.00th=[ 75], 95.00th=[ 82],
    | 99.00th=[ 113], 99.50th=[ 128], 99.90th=[ 178], 99.95th=[ 198],
    | 99.99th=[ 416]
  bw ( KiB/s): min=64680, max=64680, per=100.00%, avg=64680.00, stdev= 0.00, samples=1
  iops       : min=16170, max=16170, avg=16170.00, stdev= 0.00, samples=1
  lat (usec) : 100=98.07%, 250=1.90%, 500=0.02%, 750=0.01%
  cpu        : usr=4.60%, sys=14.01%, ctx=16384, majf=0, minf=11
  IO depths  : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwts: total=16384,0,0,0 short=0,0,0,0 dropped=0,0,0,0
    latency   : target=0, window=0, percentile=100.00%, depth=1
nonbuffered_randread: (groupid=3, jobs=1): err= 0: pid=305856: Fri Apr 11 17:07:11 2025
read: IOPS=14.0k, BW=54.8MiB/s (57.5MB/s)(64.0MiB/1168msec)
  clat (usec): min=49, max=4564, avg=68.53, stdev=38.26
    lat (usec): min=49, max=4565, avg=68.61, stdev=38.27
  clat percentiles (usec):
    | 1.00th=[ 51], 5.00th=[ 52], 10.00th=[ 55], 20.00th=[ 62],
    | 30.00th=[ 71], 40.00th=[ 71], 50.00th=[ 71], 60.00th=[ 72],
    | 70.00th=[ 72], 80.00th=[ 72], 90.00th=[ 74], 95.00th=[ 75],
    | 99.00th=[ 86], 99.50th=[ 98], 99.90th=[ 149], 99.95th=[ 165],
    | 99.99th=[ 1336]
  bw ( KiB/s): min=53320, max=57192, per=98.48%, avg=55256.00, stdev=2737.92, samples=2
  iops       : min=13330, max=14298, avg=13814.00, stdev=684.48, samples=2
  lat (usec) : 50=0.09%, 100=99.53%, 250=0.36%
  lat (msec) : 2=0.01%, 10=0.01%
  cpu        : usr=4.80%, sys=21.25%, ctx=16387, majf=0, minf=13
  IO depths  : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwts: total=16384,0,0,0 short=0,0,0,0 dropped=0,0,0,0
    latency   : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=1488MiB/s (1561MB/s), 1488MiB/s-1488MiB/s (1561MB/s-1561MB/s), io=64.0MiB (67.1MB), run=43-43msec

Run status group 1 (all jobs):
  READ: bw=222MiB/s (233MB/s), 222MiB/s-222MiB/s (233MB/s-233MB/s), io=64.0MiB (67.1MB), run=288-288msec

Run status group 2 (all jobs):
  READ: bw=62.6MiB/s (65.7MB/s), 62.6MiB/s-62.6MiB/s (65.7MB/s-65.7MB/s), io=64.0MiB (67.1MB), run=1022-1022msec

Run status group 3 (all jobs):
  READ: bw=54.8MiB/s (57.5MB/s), 54.8MiB/s-54.8MiB/s (57.5MB/s-57.5MB/s), io=64.0MiB (67.1MB), run=1168-1168msec

Disk stats (read/write):
nvme0n1: ios=46078/69, merge=1/3, ticks=1816/4, in_queue=1821, util=56.56%

```

【Q5】

Is there any bandwidth trend between these jobs? Why or why not? Please justify your answer in brief. You can also experiment with the size of the file.

■ 實驗結果摘要：

Offset (%)	Bandwidth (MiB/s)
80%	2077
60%	1896
40%	2291
20%	2354
0%	2415

■ 分析：

從上表可以觀察出，讀取位置愈接近檔案開頭（offset 越小），Bandwidth 越高，呈現出明顯的趨勢。

可能原因如下：

1. 儲存裝置的實體結構：

在傳統硬碟（HDD）中，磁碟的外圈（通常對應 offset=0%）資料密度較高，讀取速率也較快。即使是 SSD，有些控制器仍會將較快的儲存單元配置在前段，因此靠近開頭區域的讀取速度也可能更快。

2. 內部管理策略：

SSD 控制器可能在不同區段實施不同的 wear leveling、垃圾回收或 load balancing 機制，導致不同 offset 的存取效能略有差異。

3. Page cache 無干擾：

本實驗設定 `direct=1`，繞過 page cache，因此數據能夠反映儲存裝置本身的實體表現，而非快取影響。

本次實驗中可見，offset 愈小，Bandwidth 愈高，證實儲存設備的不同 LBA（邏輯區塊位址）區段的讀取效能可能不同。因此在設計系統時，可以優先將效能敏感資料配置於開頭區段，以取得較佳效能。

```
shihcw@shihcw-GF65-Thin-950: ~/Work_space/HSS/FIO/q5$ cat output_q5.txt
job1: (g=0): rw=read, bs=(R) 1024KiB-1024KiB, (W) 1024KiB-1024KiB, (T) 1024KiB-1024KiB, ioengine=psync, iodepth=1
job2: (g=1): rw=read, bs=(R) 1024KiB-1024KiB, (W) 1024KiB-1024KiB, (T) 1024KiB-1024KiB, ioengine=psync, iodepth=1
job3: (g=2): rw=read, bs=(R) 1024KiB-1024KiB, (W) 1024KiB-1024KiB, (T) 1024KiB-1024KiB, ioengine=psync, iodepth=1
job4: (g=3): rw=read, bs=(R) 1024KiB-1024KiB, (W) 1024KiB-1024KiB, (T) 1024KiB-1024KiB, ioengine=psync, iodepth=1
job5: (g=4): rw=read, bs=(R) 1024KiB-1024KiB, (W) 1024KiB-1024KiB, (T) 1024KiB-1024KiB, ioengine=psync, iodepth=1
fio-3.28
Starting 5 processes
job1: Laying out IO file (1 file / 1024MiB)
job2: Laying out IO file (1 file / 1639MiB)
job3: Laying out IO file (1 file / 1680MiB)
```

```
job1: (groupid=0, jobs=1): err= 0: pid=306152: Fri Apr 11 17:17:43 2025
read: IOPS=2077, BW=2077MiB/s (2178MB/s)(1024MiB/493msec)
    clat (usec): min=342, max=1182, avg=415.74, stdev=55.15
    lat (usec): min=342, max=1183, avg=415.85, stdev=55.17
    clat percentiles (usec):
    | 1.00th=[ 363], 5.00th=[ 371], 10.00th=[ 379], 20.00th=[ 388],
    | 30.00th=[ 392], 40.00th=[ 400], 50.00th=[ 404], 60.00th=[ 408],
    | 70.00th=[ 420], 80.00th=[ 441], 90.00th=[ 453], 95.00th=[ 494],
    | 99.00th=[ 668], 99.50th=[ 766], 99.90th=[ 988], 99.95th=[ 1188],
    | 99.99th=[ 1188]
    lat (usec)   : 500=95.80%, 750=3.52%, 1000=0.59%
    lat (msec)   : 2=0.10%
    cpu          : usr=0.20%, sys=46.75%, ctx=1037, majf=0, minf=267
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwts: total=1024,0,0,0 short=0,0,0,0 dropped=0,0,0,0
    latency     : target=0, window=0, percentile=100.00%, depth=1
```

```
job2: (groupid=1, jobs=1): err= 0: pid=306153: Fri Apr 11 17:17:43 2025
read: IOPS=1896, BW=1896MiB/s (1988MB/s)(1024MiB/540msec)
    clat (usec): min=347, max=1264, avg=457.47, stdev=71.13
    lat (usec): min=347, max=1264, avg=457.69, stdev=71.20
    clat percentiles (usec):
    | 1.00th=[ 359], 5.00th=[ 363], 10.00th=[ 371], 20.00th=[ 408],
    | 30.00th=[ 424], 40.00th=[ 429], 50.00th=[ 453], 60.00th=[ 461],
    | 70.00th=[ 482], 80.00th=[ 515], 90.00th=[ 537], 95.00th=[ 562],
    | 99.00th=[ 652], 99.50th=[ 685], 99.90th=[ 922], 99.95th=[ 1270],
    | 99.99th=[ 1270]
    lat (usec)   : 500=74.02%, 750=25.68%, 1000=0.20%
    lat (msec)   : 2=0.10%
    cpu          : usr=1.30%, sys=38.22%, ctx=1031, majf=0, minf=268
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwts: total=1024,0,0,0 short=0,0,0,0 dropped=0,0,0,0
    latency     : target=0, window=0, percentile=100.00%, depth=1
```

```
job3: (groupid=2, jobs=1): err= 0: pid=306154: Fri Apr 11 17:17:43 2025
read: IOPS=2290, BW=2291MiB/s (2402MB/s)(1024MiB/447msec)
    clat (usec): min=351, max=2290, avg=434.14, stdev=83.39
    lat (usec): min=351, max=2291, avg=434.23, stdev=83.42
    clat percentiles (usec):
    | 1.00th=[ 363], 5.00th=[ 367], 10.00th=[ 371], 20.00th=[ 383],
    | 30.00th=[ 396], 40.00th=[ 408], 50.00th=[ 416], 60.00th=[ 429],
    | 70.00th=[ 457], 80.00th=[ 478], 90.00th=[ 510], 95.00th=[ 537],
    | 99.00th=[ 627], 99.50th=[ 676], 99.90th=[ 1074], 99.95th=[ 2278],
    | 99.99th=[ 2278]
    lat (usec)   : 500=88.28%, 750=11.52%
    lat (msec)   : 2=0.10%, 4=0.10%
    cpu          : usr=0.22%, sys=28.70%, ctx=1024, majf=0, minf=268
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwts: total=1024,0,0,0 short=0,0,0,0 dropped=0,0,0,0
    latency     : target=0, window=0, percentile=100.00%, depth=1
```

```
job4: (groupid=3, jobs=1): err= 0: pid=306155: Fri Apr 11 17:17:43 2025
read: IOPS=2354, BW=2354MiB/s (2468MB/s)(1024MiB/435msec)
  clat (usec): min=358, max=2311, avg=423.01, stdev=84.48
    lat (usec): min=358, max=2312, avg=423.12, stdev=84.51
  clat percentiles (usec):
    | 1.00th=[ 367], 5.00th=[ 371], 10.00th=[ 375], 20.00th=[ 383],
    | 30.00th=[ 388], 40.00th=[ 396], 50.00th=[ 404], 60.00th=[ 420],
    | 70.00th=[ 429], 80.00th=[ 453], 90.00th=[ 482], 95.00th=[ 506],
    | 99.00th=[ 701], 99.50th=[ 766], 99.90th=[ 1090], 99.95th=[ 2311],
    | 99.99th=[ 2311]
  lat (usec)   : 500=93.85%, 750=5.57%, 1000=0.29%
  lat (msec)   : 2=0.20%, 4=0.10%
  cpu          : usr=0.46%, sys=32.72%, ctx=1024, majf=0, minf=267
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwts: total=1024,0,0,0 short=0,0,0,0 dropped=0,0,0,0
    latency    : target=0, window=0, percentile=100.00%, depth=1
```

```
job5: (groupid=4, jobs=1): err= 0: pid=306156: Fri Apr 11 17:17:43 2025
read: IOPS=2415, BW=2415MiB/s (2532MB/s)(1024MiB/424msec)
  clat (usec): min=347, max=2279, avg=412.41, stdev=79.05
    lat (usec): min=347, max=2280, avg=412.51, stdev=79.08
  clat percentiles (usec):
    | 1.00th=[ 363], 5.00th=[ 367], 10.00th=[ 371], 20.00th=[ 379],
    | 30.00th=[ 388], 40.00th=[ 396], 50.00th=[ 400], 60.00th=[ 408],
    | 70.00th=[ 412], 80.00th=[ 424], 90.00th=[ 453], 95.00th=[ 498],
    | 99.00th=[ 652], 99.50th=[ 775], 99.90th=[ 955], 99.95th=[ 2278],
    | 99.99th=[ 2278]
  lat (usec)   : 500=95.12%, 750=4.20%, 1000=0.59%
  lat (msec)   : 4=0.10%
  cpu          : usr=0.95%, sys=35.22%, ctx=1024, majf=0, minf=268
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwts: total=1024,0,0,0 short=0,0,0,0 dropped=0,0,0,0
    latency    : target=0, window=0, percentile=100.00%, depth=1
```

```
Run status group 0 (all jobs):
  READ: bw=2077MiB/s (2178MB/s), 2077MiB/s-2077MiB/s (2178MB/s-2178MB/s), io=1024MiB (1074MB), run=493-493msec

Run status group 1 (all jobs):
  READ: bw=1896MiB/s (1988MB/s), 1896MiB/s-1896MiB/s (1988MB/s-1988MB/s), io=1024MiB (1074MB), run=540-540msec

Run status group 2 (all jobs):
  READ: bw=2291MiB/s (2402MB/s), 2291MiB/s-2291MiB/s (2402MB/s-2402MB/s), io=1024MiB (1074MB), run=447-447msec

Run status group 3 (all jobs):
  READ: bw=2354MiB/s (2468MB/s), 2354MiB/s-2354MiB/s (2468MB/s-2468MB/s), io=1024MiB (1074MB), run=435-435msec

Run status group 4 (all jobs):
  READ: bw=2415MiB/s (2532MB/s), 2415MiB/s-2415MiB/s (2532MB/s-2532MB/s), io=1024MiB (1074MB), run=424-424msec

Disk stats (read/write):
nvme0n1: ios=38420/4, merge=4/0, ticks=7090/1, in_queue=7091, util=45.99%
```

【Q6】

1. (10%) Is there any significant difference between 4k and 1k read? Why or why not? Please justify your answer in brief.
2. (5%) If you want to achieve the best performance in the condition above, how would you modify blocksize? Explain it briefly.

1. Q6-1：效能比較

■ 測試結果摘要：

[4K Read]

- IOPS：58.1k
- Bandwidth：227 MiB/s
- Lat(avg)：16.80 μ s

[1K Read]

- IOPS：59.7k
- Bandwidth：58.3 MiB/s
- Lat(avg)：16.34 μ s

■ 分析：

雖然 1KB block size 的 IOPS 略高（因為資料量小、請求數量多），但整體 Bandwidth 明顯低於 4KB。

原因如下：

1. 每次 I/O 傳輸量小：1KB 一次只處理少量資料，即使 IOPS 高，也無法堆出足夠的 bandwidth。
2. 系統負擔大：1KB 每秒需處理 65,000+ 次讀取，CPU 與 I/O 系統的 context switch 成本增加。
3. 4KB 對齊：多數 SSD 與作業系統的 page 大小為 4KB，4KB block size 更適合底層儲存系統對齊設計。

因此，4KB 效能整體較佳，特別在追求 bandwidth 時更為適合。

2. Q6-2：如何調整 blocksize 以達最佳效能？

■ 回答：

若目標為最大 Bandwidth，建議調整 `bs` 至更大值，例如：`bs=64k` 或 `bs=1m`

■ 說明：

單次傳輸量大 → 可減少 I/O 次數，降低系統與磁碟控制器負擔。

順序讀寫的情境下，使用大 block size 效能明顯提升。

但在隨機小資料存取情境中，block size 過大可能導致資源浪費與延遲上升，因此仍需依應用場景調整。

```

shihcw@shihcw-GF65-Thin-9SD:~/Work_space/MSS/FIO/q6$ cat output_q6.txt
read_4k: (g=0): rw=read, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
read_1k: (g=1): rw=read, bs=(R) 1024B-1024B, (W) 1024B-1024B, (T) 1024B-1024B, ioengine=psync, iodepth=1
fio-3.28
Starting 2 processes
read_4k: Laying out IO file (1 file / 64MiB)

read_4k: (groupid=0, jobs=1): err= 0: pid=306323: Fri Apr 11 17:23:18 2025
read: IOPS=58.1k, BW=227MiB/s (238MB/s)(64.0MiB/282msec)
    clat (usec): min=13, max=129, avg=16.76, stdev= 4.86
    lat (usec): min=13, max=129, avg=16.80, stdev= 4.86
    clat percentiles (usec):
    | 1.00th=[ 15], 5.00th=[ 15], 10.00th=[ 15], 20.00th=[ 15],
    | 30.00th=[ 15], 40.00th=[ 15], 50.00th=[ 16], 60.00th=[ 17],
    | 70.00th=[ 17], 80.00th=[ 18], 90.00th=[ 19], 95.00th=[ 23],
    | 99.00th=[ 38], 99.50th=[ 46], 99.90th=[ 74], 99.95th=[ 90],
    | 99.99th=[ 124]
    lat (usec) : 20=91.83%, 50=7.78%, 100=0.37%, 250=0.02%
    cpu       : usr=5.69%, sys=28.83%, ctx=16381, majf=0, minf=13
    IO depths : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwts: total=16384,0,0,0 short=0,0,0,0 dropped=0,0,0,0
    latency   : target=0, window=0, percentile=100.00%, depth=1
read_1k: (groupid=1, jobs=1): err= 0: pid=306324: Fri Apr 11 17:23:18 2025
read: IOPS=59.7k, BW=58.3MiB/s (61.2MB/s)(64.0MiB/1097msec)
    clat (usec): min=11, max=1112, avg=16.29, stdev=11.17
    lat (usec): min=11, max=1112, avg=16.34, stdev=11.18
    clat percentiles (usec):
    | 1.00th=[ 14], 5.00th=[ 14], 10.00th=[ 14], 20.00th=[ 14],
    | 30.00th=[ 14], 40.00th=[ 15], 50.00th=[ 16], 60.00th=[ 16],
    | 70.00th=[ 17], 80.00th=[ 17], 90.00th=[ 21], 95.00th=[ 23],
    | 99.00th=[ 35], 99.50th=[ 45], 99.90th=[ 73], 99.95th=[ 88],
    | 99.99th=[ 750]
    bw ( KiB/s): min=60126, max=60370, per=100.00%, avg=60248.00, stdev=172.53, samples=2
    iops       : min=60126, max=60370, avg=60248.00, stdev=172.53, samples=2
    lat (usec) : 20=88.39%, 50=11.26%, 100=0.31%, 250=0.03%, 1000=0.01%
    lat (msec)  : 2=0.01%
    cpu       : usr=7.39%, sys=30.75%, ctx=65528, majf=0, minf=14
    IO depths : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwts: total=65536,0,0,0 short=0,0,0,0 dropped=0,0,0,0
    latency   : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=227MiB/s (238MB/s), 227MiB/s-227MiB/s (238MB/s-238MB/s), io=64.0MiB (67.1MB), run=282-282msec

Run status group 1 (all jobs):
  READ: bw=58.3MiB/s (61.2MB/s), 58.3MiB/s-58.3MiB/s (61.2MB/s-61.2MB/s), io=64.0MiB (67.1MB), run=1097-1097msec

Disk stats (read/write):
nvme0n1: ios=77791/29, merge=0/19, ticks=913/17, in_queue=938, util=56.52%

```

【Q7】

Please explain how you achieve the fastest 1G nonbuffered read in brief.

■ 思考方向：

- Block size 越大，bandwidth 越高（Q6 證實）
- 順序讀比隨機讀效能高很多（Q1 證實）
- direct=1 能繞過 cache，但會真實測試儲存效能
- 使用單一大檔案進行連續讀取，配合大 block size 是最快方式

■ 測試結果摘要：

[Fastest_1G_non-buffered_read]

- IOPS：1943
- Bandwidth：1943 MiB/s (2037 MB/s)
- Lat(avg)：512.72 μ s

■ 分析：

本次測試使用了大區塊大小（1 MiB），並選擇順序讀取與 non-buffered 模式，有效減少了每次操作的延遲與 CPU 負擔，並成功達到超過 1900 MiB/s 的傳輸速度。這驗證了前面的思考方向：

1. 大區塊大小（如 1 MiB）對提升儲存效能有顯著影響，特別是在順序讀取的情況下。
2. non-buffered 模式減少了快取的影響，使得效能能夠接近儲存設備的最大讀取能力。因此，這樣的配置達到了最優的讀取效能。

```
shihcw@shihcw-GF65-Thin-950:~/Work_space/HSS/FIO/q7$ cat output_q7.txt
fastest_1g_read: (g=0): rw=read, bs=(R) 1024KiB-1024KiB, (W) 1024KiB-1024KiB, (T) 1024KiB-1024KiB, ioengine=psync, iodepth=1
fio-3.28
Starting 1 process
fastest_1g_read: Laying out IO file (1 file / 1024MiB)

fastest_1g_read: (groupid=0, jobs=1): err= 0: pid=306539: Fri Apr 11 17:29:38 2025
read: IOPS=2420, BW=2421MiB/s (2538MB/s)(1024MiB/423msec)
   clat (usec): min=353, max=960, avg=412.00, stdev=38.02
   lat (usec): min=353, max=960, avg=412.11, stdev=38.03
   clat percentiles (usec):
   |  1.00th=[ 363],  5.00th=[ 371], 10.00th=[ 379], 20.00th=[ 392],
   | 30.00th=[ 400], 40.00th=[ 404], 50.00th=[ 408], 60.00th=[ 412],
   | 70.00th=[ 416], 80.00th=[ 420], 90.00th=[ 445], 95.00th=[ 461],
   | 99.00th=[ 537], 99.50th=[ 594], 99.90th=[ 766], 99.95th=[ 963],
   | 99.99th=[ 963]
   lat (usec)  : 500=97.95%, 750=1.76%, 1000=0.29%
   cpu         : usr=0.71%, sys=22.51%, ctx=1024, majf=0, minf=267
   IO depths   : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
   submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
   complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
   issued rwts: total=1024,0,0,0 short=0,0,0,0 dropped=0,0,0,0
   latency    : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
   READ: bw=2421MiB/s (2538MB/s), 2421MiB/s-2421MiB/s (2538MB/s-2538MB/s), io=1024MiB (1074MB), run=423-423msec

Disk stats (read/write):
nvme0n1: ios=6085/0, merge=0/0, ticks=1179/0, in_queue=1179, util=61.69%
```

【Bonus】(10%)

1. (5%) Compare with multiple kinds of storage devices

Is there any significant difference between read/randread on storage devices? Why or why not? Please justify your answer in brief.

2. (5%) Find the specs of your own hardware that you tested on

Is your hardware running to spec? If not, could you come up with a possible theory?

1. Bonus-1：與多種儲存裝置進行比較

■ 測試裝置：

裝置名稱：Kingston DataTraveler EXODIA ONYX

裝置類型：USB Flash Drive

容量版本：128GB

接頭介面：USB 3.2 Gen 1（向下相容 USB 2.0）

理論速度：USB 3.2 Gen 1（註明「速度會因裝置與使用情況而異」）

尺寸重量：60.7mm × 21mm × 10.2mm，重約 8g

作業系統支援：Windows / macOS / Linux / Chrome OS

測試檔案：testfile_usb（64MiB）

測試工具：FIO

規格表	
儲存容量 ¹	64GB 128GB 256GB
傳輸介面	USB 3.2 Gen 1
速度 ¹	USB 3.2 Gen 1
尺寸	60.7mm x 21mm x 10.2mm
重量	8g
作業溫度	0°C–60°C
儲存溫度	-20°C–85°C
保固	5 年保固，免費技術支援
相容於	Windows [®] 11, 10, macOS [®] (v.10.15.x+), Linux (v. 4.4+), Chrome OS [™]
¹ 速度可能因主機及裝置設定而異。 ² 快閃儲存裝置的標示容量中，有部分用於格式化和其他功能，並非全部用於資料儲存。因此，實際可用來儲存資料的儲存容量會略低於產品所標示之容量。如需更多詳細資訊，請參閱 Kingston《快閃記憶體指南》，網址：kingston.com/TestGuide	

■ 測試結果摘要：

[Read]

- IOPS：2291
- Bandwidth：11.7MiB/s (12.3 MB/s)
- Lat(avg)：333.53 μs

[Randread]

- IOPS：1761
- Bandwidth：7045KiB/s (7.2 MB/s)
- Lat(avg)：556.04 μs

2. Bonus-1：

Is there any significant difference between read/randread on storage devices? Why or why not? Please justify your answer in brief.

Ans：

1. read 明顯優於 randread：

順序讀取（read）帶寬高出約 70%，IOPS 也明顯較高。

隨機讀取需要較多位置跳躍與地址查詢，在 USB 上會造成延遲累積。

2. 原因分析：

USB 隨身碟設計本身較簡單，無法有效處理大量隨機讀取請求。

控制器沒有 DRAM 快取與複雜調度機制，面對 `randread` 顯得吃力。

USB 接口本身（尤其若為 USB 2.0）帶寬較小也限制其表現。

3. 觀察：

clat 延遲方面，`randread` 幾乎是 `read` 的 1.7 倍。

randread 約有 76.75% 的操作都集中在 750μs 延遲區段，明顯拉高整體平均。

3. Bonus-2：找到您測試過的硬體的規格並核對

■ 測試裝置：

裝置名稱：Kingston DataTraveler EXODIA ONYX

裝置類型：USB Flash Drive

容量版本：128GB

接頭介面：USB 3.2 Gen 1（向下相容 USB 2.0）

理論速度：USB 3.2 Gen 1（註明「速度會因裝置與使用情況而異」）

尺寸重量：60.7mm × 21mm × 10.2mm，重約 8g

作業系統支援：Windows / macOS / Linux / Chrome OS

測試檔案：testfile_usb（64MiB）

測試工具：FIO

規格表

儲存容量 ¹
64GB 128GB 256GB
傳輸介面
USB 3.2 Gen 1
速度 ¹
USB 3.2 Gen 1
尺寸
60.7mm x 21mm x 10.2mm
重量
8g
作業溫度
0°C~60°C
儲存溫度
-20°C~85°C
保固
5 年保固，免費技術支援
相容於
Windows® 11, 10, macOS® (v.10.15.x+), Linux (v. 4.4+), Chrome OS™

¹ 速度可能因主機及裝置設定而異。

² 快閃儲存裝置的標示容量中，有部分用於格式化和其他功能，並非全部用於資料儲存。因此，實際可用來儲存資料的儲存容量會低於產品所標示之容量。如需更多詳細資訊，請參閱 Kingston《快閃記憶體指南》，網址：kingston.com/testinguide。

4. Bonus-2 :

Is your hardware running to spec? If not, could you come up with a possible theory?

Ans :

- 順序讀取 read 的 BW 約 12.3 MB/s(僅用 FIO + 64MiB 測試檔)
- 根據 USB 3.2 Gen 1 理論帶寬 (約 5 Gbps $\approx$ 625 MB/s)，但實際 USB 隨身碟常見效能遠低於此，特別是入門型號多落在 10–40 MB/s 讀取範圍內。
- 此裝置的 12.3 MB/s 讀取速度符合入門 USB 3.2 Gen 1 隨身碟的水準，並未明顯低於規格期待，因此可算是正常發揮。
- 測得順序讀取約 12.3MB/s，雖未公布官方實際速度，但與預期相符。
- 隨機讀取則因 USB 隨身碟控制器與快取結構限制，效能下降為合理現象。
- 裝置運行狀況正常，未見異常降速情形，推估硬體表現屬於設計範圍內。

```
shihcw@shihcw-GF65-Thin-950:/media/shihcw/KINGSTON/HSS_bonus$ cat output_usb.txt
read_usb: (g=0): rw=read, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
randread_usb: (g=1): rw=randread, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T) 4096B-4096B, ioengine=psync, iodepth=1
fio-3.28
Starting 2 processes
read_usb: Laying out IO file (1 file / 64MiB)

read_usb: (groupid=0, jobs=1): err= 0: pid=311011: Fri Apr 11 18:43:55 2025
  read: IOPS=2991, BW=11.7MiB/s (12.3MB/s)(64.0MiB/5476msec)
    clat (usec): min=258, max=1759, avg=333.53, stdev=42.25
    lat (usec): min=258, max=1759, avg=333.59, stdev=42.26
    clat percentiles (usec):
      | 1.00th=[ 273],  5.00th=[ 281], 10.00th=[ 293], 20.00th=[ 302],
      | 30.00th=[ 310], 40.00th=[ 322], 50.00th=[ 330], 60.00th=[ 343],
      | 70.00th=[ 351], 80.00th=[ 367], 90.00th=[ 379], 95.00th=[ 388],
      | 99.00th=[ 404], 99.50th=[ 424], 99.90th=[ 570], 99.95th=[ 693],
      | 99.99th=[ 1631]
    bw ( KIB/s): min=11802, max=12144, per=100.00%, avg=11973.70, stdev=98.57, samples=10
    iops       : min= 2950, max= 3036, avg=2993.30, stdev=24.77, samples=10
    lat (usec) : 500=99.75%, 750=0.20%, 1000=0.01%
    lat (msec) : 2=0.04%
    cpu        : usr=0.68%, sys=2.14%, ctx=16384, majf=0, minf=12
    IO depths  : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwts: total=16384,0,0,0 short=0,0,0,0 dropped=0,0,0,0
    latency    : target=0, window=0, percentile=100.00%, depth=1

randread_usb: (groupid=1, jobs=1): err= 0: pid=311012: Fri Apr 11 18:43:55 2025
  read: IOPS=1761, BW=7045KiB/s (7214kB/s)(64.0MiB/9303msec)
    clat (usec): min=264, max=1114, avg=566.04, stdev=122.19
    lat (usec): min=264, max=1115, avg=566.19, stdev=122.23
    clat percentiles (usec):
      | 1.00th=[ 297],  5.00th=[ 322], 10.00th=[ 355], 20.00th=[ 506],
      | 30.00th=[ 570], 40.00th=[ 578], 50.00th=[ 594], 60.00th=[ 611],
      | 70.00th=[ 627], 80.00th=[ 635], 90.00th=[ 652], 95.00th=[ 701],
      | 99.00th=[ 889], 99.50th=[ 988], 99.90th=[ 1012], 99.95th=[ 1037],
      | 99.99th=[ 1090]
    bw ( KIB/s): min= 5760, max= 7384, per=99.18%, avg=6987.11, stdev=410.36, samples=18
    iops       : min= 1440, max= 1846, avg=1746.78, stdev=102.59, samples=18
    lat (usec) : 500=19.96%, 750=76.75%, 1000=3.09%
    lat (msec) : 2=0.20%
    cpu        : usr=0.77%, sys=2.19%, ctx=16385, majf=0, minf=12
    IO depths  : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwts: total=16384,0,0,0 short=0,0,0,0 dropped=0,0,0,0
    latency    : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=11.7MiB/s (12.3MB/s), 11.7MiB/s-11.7MiB/s (12.3MB/s-12.3MB/s), io=64.0MiB (67.1MB), run=5476-5476msec

Run status group 1 (all jobs):
  READ: bw=7045KiB/s (7214kB/s), 7045KiB/s-7045KiB/s (7214kB/s-7214kB/s), io=64.0MiB (67.1MB), run=9303-9303msec

Disk stats (read/write):
  sda: ios=32311/0, merge=0/0, ticks=13550/0, in_queue=13550, util=91.87%
```